

On the Complexity of Computing with p -adic Lattices

Pierre Fromonot
Université de Limoges, Limoges
France
pierre.fromonot@etu.unilim.fr

Mercedes Haiech
Université de Limoges, CNRS, XLIM
UMR 7252
Limoges, France
mercedes.haiech@unilim.fr

Tristan Vaccon
Université de Limoges, CNRS, XLIM
UMR 7252
Limoges, France
tristan.vaccon@unilim.fr

Abstract

We study the complexity of two fundamental problems attached to p -adic lattices: computing an N -orthogonal basis, and deciding whether a given basis is N -orthogonal. In the black-box model, where the ultrametric norm N is fixed and only accessible through an evaluation oracle, we prove that both problems require $\Omega(p^{v-1})$ queries in the worst case, where $v \leq n$ is the largest multiplicity of a commensurability class of N and n is the dimension of the ambient vector space. We then analyse algorithms that reach these bounds up to a polynomial factor, which proves their optimality. By contrast, when the norm is supplied through an explicit diagonal representation, both problems become polynomial.

Our results answer formally an open question of Zhang, Deng and Wang [13] and refine the recent results of Duan and Wu [4] by providing pointwise query bounds depending on the norm's structure, in addition to uniform bounds.

CCS Concepts

• Computing methodologies → Algebraic algorithms.

Keywords

p -adic lattices, ultrametric norms, orthogonalisation, query complexity, LVP, CVP, post-quantum cryptography

ACM Reference Format:

Pierre Fromonot, Mercedes Haiech, and Tristan Vaccon. 2027. On the Complexity of Computing with p -adic Lattices. In *52nd International Symposium on Symbolic and Algebraic Computation (ISSAC '27)*, July 12–16, 2027, Tsukuba, Japan. ACM, New York, NY, USA, 9 pages. <https://doi.org/XX.XXXX/XXXXXXXXXXXXXXX>

1 Introduction

1.1 History and related results

In 2018, Deng, Luo and Xiao [3] introduced the Longest Vector Problem (LVP) and the Closest Vector Problem (CVP) for p -adic lattices, as analogues of the Euclidean SVP (Shortest Vector Problem) and CVP on which modern post-quantum cryptography is built (e.g. Kyber, Dilithium). In such spaces, defined over a finite extension $K/\mathbb{Q}_p$, the norm is ultrametric, which gives p -adic lattices a geometry radically different from that of their Euclidean counterparts and opens the way to new cryptographic hardness assumptions [6].

These problems were then generalised to p -adic lattices embedded in $\mathbb{Q}_p$ -vector spaces of finite dimension n , equipped with an arbitrary ultrametric norm N [12]. Zhang, Deng and Wang proved that LVP and CVP are solvable in polynomial time as soon as one is given an N -orthogonal basis, a structure whose existence is always guaranteed in ultrametric geometry [10], and that CVP and the computation of such a basis are polynomially equivalent.

More recently, Zhang et al. [13] showed that the usual norm $|\cdot|_p$ on a field extension $K/\mathbb{Q}_p$ reveals a uniformiser, from which an orthogonal basis can be built efficiently: this made LVP and CVP trivial in that setting, and completely broke the first cryptosystem based on these structures [2]. Recalling that every ultrametric norm admits a “diagonal” form, out of which an N -orthogonal basis can be read off immediately, the authors raised the following fundamental question: *is it possible to find a diagonal form (or an orthogonal basis) of a given norm when one only has access to an evaluation oracle?*

This paper answers that question formally: one can compute an orthogonal basis, with at least exponentially many queries in the worst case.

Independently and very recently, Duan and Wu [4] studied the same norm-query model. They prove that orthogonalisation admits no uniform query bound depending only on the dimension: for every deterministic algorithm that outputs an N -orthogonal basis for every ultrametric norm N , the number of queries is unbounded as N varies, even at fixed dimension.

While Euclidean lattice-based cryptography dominates the post-quantum landscape, p -adic lattices offer a potential alternative should the hardness assumptions behind LWE turn out to be weakened, as suggested by the recent work of Daniel R. Simon [9].

1.2 Contributions and organisation

The first observation is that p -adic commensurability—the equivalence relation $v_1 \sim v_2$ when $N(v_1)/N(v_2) \in p^{\mathbb{Z}}$ —is the right parameter governing the difficulty of the problems considered here: orthogonalization and orthogonality decision. In Section 2 we introduce this notion. Section 3 is devoted to the study of the explicit family $N_{P(\mathbb{F}_p^n)}$ of $\frac{p^n-1}{p-1}$ pairwise hardly distinguishable norms.

Let $v_{\max} \leq n$ be the largest multiplicity of a commensurability class of N . The hardest case for the orthogonalization problem is reached when $v_{\max} = n$. This is attained by many natural norms, including the usual sup norm and every norm in our family $N_{P(\mathbb{F}_p^n)}$. We prove in Section 4 our first main result:

THEOREM 4.1: Lower Bound Orthogonalization. *Any algorithm (deterministic, or randomised with success probability $\geq 2/3$) which reconstructs an (A, c) -representation of N from a norm oracle performs $\Omega(p^{v_{\max}-1})$ oracle calls in the worst case. In particular, for $v_{\max} = n: \Omega(p^{n-1})$.*



This work is licensed under a Creative Commons Attribution 4.0 International License. *ISSAC '27, Tsukuba, Japan*

© 2027 Copyright held by the owner/author(s).

ACM ISBN XX.XXXX/XXXXXXXXXXXXXXXXXXXX

<https://doi.org/XX.XXXX/XXXXXXXXXXXXXXX>

For the decision problem, the relevant parameter is instead the size of the largest commensurability block in the input family, denoted $l_{\max}$. In particular, for the usual sup norm and every norm in $N_{P(\mathbb{F}_p^n)}$, all nonzero vectors are commensurable. Thus, a family of m vectors has $l_{\max} = m$, and a basis $l_{\max} = n$. This leads to our second main result (Section 5):

THEOREM 5.4: Lower Bound Decision. *Any algorithm (deterministic, or randomised with success probability $\geq 2/3$) that decides whether a basis B is N -orthogonal performs $\Omega(p^{l_{\max}-1})$ oracle calls in the worst case. In particular, if $l_{\max} = n$: $\Omega(p^{n-1})$.*

We show that these bounds are tight (up to polynomial factors) by analysing explicit algorithms: Algorithm 2 for orthogonalization and Algorithm 4 for decision. In particular, in the single block case $l_{\max} = n$, our decision algorithm makes $O\left(\frac{p^n-1}{p-1}\right) = O(p^{n-1})$ oracle calls. Combined with the lower bound, we get a tight bound of $\Theta(p^{n-1})$.

Our exponential lower bounds do not result from an unbounded or degenerate family of norms. The family $N_{P(\mathbb{F}_p^n)}$ is finite, explicit, and all its members take their values in $p^{\mathbb{Z}}$. Nevertheless, distinguishing its elements already requires $\Omega(p^{n-1})$ queries. We then identify two distinct parameters that govern the cost of orthogonalization. The multiplicity $v_{\max}$ of the largest commensurability class yields an intrinsic barrier $\Omega(p^{v_{\max}-1})$ that no algorithm can avoid, whereas the geometric conditioning D of the instance (defined in Section 4.2) accounts for the failure of a uniform bound over all norms, as shown by Duan & Wu in [4]. From a cryptographic perspective, D is not a plausible security parameter. It measures the instance-dependent conditioning of the problem: it grows linearly with n and logarithmically with the ratios between the norms of the current vectors and their distances to the previously constructed subspaces, whereas $p^{v_{\max}-1}$ is the intrinsic source of hardness.

By contrast, when a diagonal representation (A, c) is given, our *residual commensurability* criterion (Section 5) decides orthogonality in $O(nm^2)$ arithmetic operations: hardness lies in the norm oracle, not only in the norm structure itself.

THEOREM 5.1: Residual Commensurability Criterion . *Let N be a norm on V and let $b_1, \dots, b_m$ be nonzero vectors of V . Let $B_1, \dots, B_k$ be their commensurability classes and $\bar{M}_1, \dots, \bar{M}_k$ the associated residual matrices of leading digits. Then the family is N -orthogonal if and only if $\ker \bar{M}_i = \{0\}$ over $\mathbb{F}_p$ for every block B_i .*

SAGEMATH implementations of all our algorithms are publicly available at : <https://github.com/Synthetikk/p-adic-Lattices-for-Cryptography>.

2 Norms and p -adic lattices

2.1 The norm-oracle model

Throughout the paper we consider a black-box norm oracle: it takes as input a vector in $\mathbb{Q}_p^n$ and returns the exact value of its norm. For each fixed norm N , we assume that its values are taken in a discrete subset $\Gamma \subset \mathbb{R}$, allowing exact equality tests, order comparisons, tests for membership in $p^{\mathbb{Z}}$, and usual arithmetic operations.

We distinguish oracle-query complexity from arithmetic complexity. For algorithms operating through a norm oracle, our bounds

concern the number of oracle queries, while the associated arithmetic work is measured separately and may depend on that number of calls. Each individual arithmetic operation is assumed to have polynomial cost.

2.2 Basic facts on p -adic numbers

We recall the minimal background on p -adic numbers; more details can be found in [8]. Let p be a prime number. For $x \in \mathbb{Q}^*$, write $x = p^t \cdot \frac{a}{b}$ with $t, a, b \in \mathbb{Z}$, $b \neq 0$ and p dividing neither a nor b , and set $|x|_p = p^{-t}$, $|0|_p = 0$. Then $|\cdot|_p$ is a non-archimedean absolute value on $\mathbb{Q}$: it is multiplicative, it vanishes only at 0, and it satisfies $|x + y|_p \leq \max\{|x|_p, |y|_p\}$, with equality whenever $|x|_p \neq |y|_p$. We call this last property the *ultrametric inequality* (or the *ultrametric equality* when it is one).

We denote by $\mathbb{Q}_p$ the completion of $\mathbb{Q}$ for $|\cdot|_p$, by $\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \leq 1\}$ its ring of integers — a compact discrete valuation ring, with maximal ideal $p\mathbb{Z}_p$ and valuation v_p — and by $\mathbb{Z}_p^\times = \{x \in \mathbb{Q}_p : |x|_p = 1\}$ its group of units. For a positive integer r , we denote by $P(\mathbb{F}_p^r)$ the projective space of $\mathbb{F}_p^r$, which we simply regard as the set of one-dimensional subspaces (the vector lines) of $\mathbb{F}_p^r$. It is well-known that $\text{card}(P(\mathbb{F}_p^r)) = 1 + p + \dots + p^{r-1} = \frac{p^r-1}{p-1}$.

2.3 Norms and orthogonal bases

Since $\mathbb{Q}_p$ is a field, $\mathbb{Q}_p$ -vector spaces are defined as usual; we denote them by V , and assume throughout that $\dim_{\mathbb{Q}_p} V = n \geq 2$ is finite. A *norm* N on V is a map $V \rightarrow \mathbb{R}_{\geq 0}$ such that $N(v) = 0$ if and only if $v = 0$, $N(\alpha v) = |\alpha|_p N(v)$ for all $\alpha \in \mathbb{Q}_p$ and $v \in V$, and $N(v + w) \leq \max\{N(v), N(w)\}$ for all $v, w \in V$, with equality whenever $N(v) \neq N(w)$ — the ultrametric inequality again. Note that, by induction, it yields $N(v_1 + \dots + v_k) = \max_i N(v_i)$ as soon as the $N(v_i)$ are pairwise distinct.

PROPOSITION 2.1 (WEIL, [10]). *Let N be a norm on V . Then there is a decomposition $V = V_1 \oplus V_2 \oplus \dots \oplus V_n$ of V into a direct sum of n one-dimensional subspaces such that*

$$N\left(\sum_{i=1}^n v_i\right) = \max_{1 \leq i \leq n} N(v_i) \quad (1)$$

for all $v_i \in V_i$.

DEFINITION 2.2 (N-ORTHOGONAL BASIS, [10]). *Let N be a norm on V . A basis $(b_1, \dots, b_n)$ of V is N -orthogonal if the decomposition $V = V_1 \oplus \dots \oplus V_n$ with $V_i = \text{Vect}_{\mathbb{Q}_p}(b_i)$ satisfies (1). A family $(b_1, \dots, b_m)$ of nonzero vectors is N -orthogonal if it is an N -orthogonal basis of the subspace it spans.*

Thus Proposition 2.1 states that for any norm N , V admits an N -orthogonal basis. The directness of the sum is in fact a consequence of (1): if $\sum \alpha_i b_i = 0$ then $\alpha_i b_i = 0$ for every i . As such, proving (1) is enough to characterise orthogonality.

THEOREM 2.3 (GOLDMAN–IWAHORI, [13]). *Let $(b_1, \dots, b_n)$ be a basis of V . Then every norm N on V can be written as*

$$N\left(\sum_{i=1}^n \alpha_i b_i\right) = \max_{1 \leq r \leq n} c_r \left| \sum_{i=1}^n \alpha_i a_{r,i} \right|_p,$$

where $\alpha_i \in \mathbb{Q}_p$, $c_r \in \mathbb{R}_{>0}$ for $r \in \{1, \dots, n\}$, and $(a_{r,i})_{n \times n} \in \text{GL}_n(\mathbb{Q}_p)$.

PROOF. By Proposition 2.1 there is an N -orthogonal basis $(e_1, \dots, e_n)$ of V ; put $c_i := N(e_i)$ and let $A = (a_{r,i}) \in \text{GL}_n(\mathbb{Q}_p)$ be the change of basis matrix, $(b_1 \cdots b_n) = (e_1 \cdots e_n) \cdot A$. $\square$

We call (A, c) -representation of a norm N on V its expression in the form of Theorem 2.3, and say that N is given in *diagonal form* when it is presented by such a pair (A, c) .

The proof makes it clear that knowing an (A, c) -representation of N is the same as knowing an N -orthogonal basis: the columns of A^{-1} are the coordinates, in the basis $(b_1, \dots, b_n)$, of the vectors of an N -orthogonal basis.

Since A is invertible, its rows define a basis of the dual space $V^\vee$. We write L_r for the corresponding r -th linear form, i.e. $L_r(v) = a_{r,1}v_1 + \dots + a_{r,n}v_n$ for every $v = \sum_{i=1}^n v_i b_i \in V$. The norm then reads $N(v) = \max_{1 \leq r \leq n} c_r |L_r(v)|_p$.

EXAMPLE 2.4. Consider the norm defined, for every $v = (v_1, v_2, v_3)^T \in \mathbb{Q}_p^3$ expressed in a basis (b_1, b_2, b_3) of V , by

$$N(v) = \max(\pi |v_1|_p, \sqrt{2} |v_1 + v_2|_p, |v_1 + v_2 + v_3|_p),$$

that is, $c_1 = \pi$, $c_2 = \sqrt{2}$, $c_3 = 1$ and $L_1(v) = v_1$, $L_2(v) = v_1 + v_2$, $L_3(v) = v_1 + v_2 + v_3$, so that A is the lower triangular matrix all of whose entries on and below the diagonal are 1. The columns of A^{-1} then give $e_1 = b_1 - b_2$, $e_2 = b_2 - b_3$, $e_3 = b_3$, an N -orthogonal basis of V .

PROPOSITION 2.5. If $(e_1, \dots, e_n)$ is an N -orthogonal basis of V and $e'_i = e_i + \varepsilon_i$ with $N(\varepsilon_i) < N(e_i)$, then (e'_i) is N -orthogonal, with the same c_i 's.

PROOF. $N(\sum \alpha_i \varepsilon_i) \leq \max_i |\alpha_i|_p N(\varepsilon_i) < \max_i |\alpha_i|_p N(e_i) = N(\sum \alpha_i e_i)$, and the ultrametric equality concludes. $\square$

This is very good news: unlike in the real case, no infinite precision is needed to know orthogonality exactly. We finally record three lemmas that will be used repeatedly.

LEMMA 2.6 ([1, 11]). Let N be a norm on V . (i) A basis $(b_1, \dots, b_n)$ of V is N -orthogonal if and only if $N(\sum_{i=1}^n \alpha_i b_i) = \max_{1 \leq i \leq n} N(\alpha_i b_i)$ for all $\alpha_1, \dots, \alpha_n$ such that one of them is a unit and the other $n-1$ lie in $\mathbb{Z}_p$. (ii) A family $(b_1, \dots, b_m)$ with $N(b_1) = \dots = N(b_m) =: \rho$ is N -orthogonal if and only if $p \mid \alpha_i$ for $1 \leq i \leq m$ whenever $\alpha_1, \dots, \alpha_m \in \mathbb{Z}_p$ satisfy $N(\sum_{i=1}^m \alpha_i b_i) < \rho$.

LEMMA 2.7. Let $(b_1, \dots, b_m)$ be a family of vectors of V with $N(b_1) = \dots = N(b_m) = \rho$. Then $(b_1, \dots, b_m)$ is N -orthogonal if and only if $N(\sum_{i=1}^m \tilde{\alpha}_i b_i) = \rho$ for all $\tilde{\alpha}_i \in \{0, \dots, p-1\}$, not all zero.

PROOF. By Lemma 2.6, the only combinations that can make the norm drop are those whose scalars are all divisible by p ; so if there are $\alpha_i \in \{0, \dots, p-1\}$, not all zero, with $N(\sum \alpha_i b_i) < \rho$, the family is not N -orthogonal. Conversely, if it is not N -orthogonal, there are $\alpha_i = \alpha_i + p\mu_i \in \mathbb{Z}_p$, with $\mu_i \in \mathbb{Z}_p$ and at least one $\alpha_i \in \{0, \dots, p-1\}$ nonzero, such that $N(\sum \alpha_i b_i) < \rho$; since $N(p\mu_i b_i) \leq \rho/p < \rho$, the strict ultrametric inequality gives $N(\sum \alpha_i b_i) < \rho$. $\square$

2.4 Commensurability and orthogonality

Let N be a norm on V and let (A, c) be a diagonal representation of N . We write $\kappa_i = c_i p^{\mathbb{Z}} \in \mathbb{R}_{>0}/p^{\mathbb{Z}}$ for the classes induced by the values c_i , and call them *commensurability classes*. The set $\Gamma := \bigcup_i c_i p^{\mathbb{Z}} \cup \{0\}$ of values taken by N being discrete, the balls $B_{\leq \rho} := \{v \in V : N(v) \leq \rho\}$, which are free $\mathbb{Z}_p$ -modules, stagnate and then grow by sudden jumps as the radius ρ increases: one gets an infinite chain of strictly nested lattices, that is, a filtration, whose associated graded piece $\text{gr}_\rho := B_{\leq \rho}/B_{< \rho}$ is an $\mathbb{F}_p$ -vector space of dimension

$$\nu_\rho := \text{card}\{i : c_i \in \rho p^{\mathbb{Z}}\} = n_{\kappa(\rho)},$$

the multiplicity of the class of ρ . We write in particular $\nu_{\max} := \max_\kappa n_\kappa \in \{1, \dots, n\}$; for instance the parameters $c_1 = \pi$, $c_2 = \sqrt{2}$, $c_3 = 1$ of Example 2.4 give three distinct classes modulo $p^{\mathbb{Z}}$, so that $\nu_{\max} = 1$ there.

We are also interested in the commensurability of given vectors of V : two nonzero vectors $v_1, v_2 \in V$ are *commensurable* (with respect to N) when the ratio of their norms lies in the image of $|\cdot|_p$, that is when $\frac{N(v_1)}{N(v_2)} \in p^{\mathbb{Z}}$, and *incommensurable* otherwise. This is clearly an equivalence relation, and here is why it is relevant to orthogonality.

PROPOSITION 2.8. Let N be a norm on V . Two incommensurable vectors $v_1, v_2 \in V$ are N -orthogonal.

PROOF. Let $\alpha_1, \alpha_2 \in \mathbb{Q}_p$; if one of them vanishes, the equality $N(\alpha_1 v_1 + \alpha_2 v_2) = \max(N(\alpha_1 v_1), N(\alpha_2 v_2))$ is obvious. Otherwise $|\alpha_1/\alpha_2|_p \in p^{\mathbb{Z}}$ while $N(v_1)/N(v_2) \notin p^{\mathbb{Z}}$, so that $\frac{N(\alpha_1 v_1)}{N(\alpha_2 v_2)} \notin p^{\mathbb{Z}}$; in particular this ratio cannot equal 1, the two norms differ, and the ultrametric equality gives the claim. $\square$

REMARK 2.9. (i) The converse is false: for the usual sup norm $\|v\|_\infty = \max_{1 \leq i \leq n} |v_i|_p$ on $\mathbb{Q}_p^n$ all nonzero vectors are commensurable, yet many of them are N -orthogonal. (ii) A norm oracle is enough to determine commensurability. (iii) If the vectors of a family $(b_1, \dots, b_m)$ are pairwise incommensurable, then the family is N -orthogonal: this is the case $k = m$ of Lemma 2.10 below.

Commensurability classes split orthogonality into independent blocks, which is the key structural fact behind all the algorithms of this paper.

LEMMA 2.10 (BLOCK DECOMPOSITION). Let $b_1, \dots, b_m$ be nonzero vectors of V and let $B_1, \dots, B_k$ be their commensurability classes. Then $(b_1, \dots, b_m)$ is N -orthogonal if and only if each block B_i is N -orthogonal.

PROOF. A subfamily of an N -orthogonal family is N -orthogonal, which gives one implication. Conversely, assume that every block is N -orthogonal and let $b = \sum_{j=1}^m \alpha_j b_j$. Grouping the terms blockwise, write $b = \sum_{i=1}^k w_i$ with $w_i = \sum_{j \in B_i} \alpha_j b_j$. By orthogonality of the block B_i we have $N(w_i) = \max_{j \in B_i} N(\alpha_j b_j)$, which lies in the class κ_i of B_i whenever $w_i \neq 0$. The classes $\kappa_1, \dots, \kappa_k$ being pairwise distinct, the nonzero $N(w_i)$ are pairwise distinct, so that $N(b) = \max_{1 \leq i \leq k} N(w_i) = \max_{1 \leq j \leq m} N(\alpha_j b_j)$. $\square$

Given a family B of vectors of V , we write $B_1, \dots, B_k$ ($k \leq m$) for the commensurability classes of its vectors, l_i for the number of vectors in B_i , and $l_{\max} := \max_i \text{card } B_i \geq \nu_{\max}$. These classes can be

Algorithm 1: Commensurability classes

Input: m vectors $b_1, \dots, b_m$ of V , a norm oracle for N

Output: the list of the commensurability classes

```

1 precompute the norms  $N(b_j)$ ,  $1 \leq j \leq m$ ;
2  $Classes \leftarrow []$ ;  $rest \leftarrow (b_1, \dots, b_m)$ ;
3 while  $rest \neq []$  do
4   pick  $b_0 \in rest$  and let  $C$  be the set of the  $b \in rest$  such
     that  $N(b)/N(b_0) = p^{k_b} \in p^{\mathbb{Z}}$ ;
5   append  $\{p^{k_b}b : b \in C\}$  to  $Classes$ ; // all of norm
      $N(b_0)$ 
6    $rest \leftarrow rest \setminus C$ ;
7 return  $Classes$ ;
```

computed from the oracle alone, by Algorithm 1 above; they span subspaces which are automatically N -orthogonal to one another, and by Lemma 2.10 nothing else has to be checked outside the blocks. Inside a block, however, nothing is known yet: one has to refine the criterion and look for a cancellation of the leading terms in a linear combination $\alpha_1 v_1 + \alpha_2 v_2$. Methods for this are given in Section 5.

Algorithm 1 computes the commensurability classes of a family B of m vectors in m oracle queries and at most $\binom{m}{2} = O(m^2)$ divisions. In the most expensive case, all the vectors of B are pairwise incommensurable, and thus, one gets that B is orthogonal for free.

EXAMPLE 2.11. Take again the norm $N(v) = \max(\pi|v_1|_p, \sqrt{2}|v_1 + v_2|_p, |v_1 + v_2 + v_3|_p)$ of Example 2.4 and consider the basis

$$B = \begin{pmatrix} 1 & 1+p^2 & p^{-2} \\ p^4 & p^3 & 1 \\ p^{-2}+p & p+p^2 & 1 \end{pmatrix},$$

whose vectors are the columns. One has $N(b_1) = p^2$, $N(b_2) = \pi$ and $N(b_3) = \pi p^2$. As such, the commensurability classes are $\{b_1\}$ and $\{b_2, b_3\}$ and $l_{\max} = 2$. One can normalize b_3 by replacing it with $b'_3 = p^2 b_3 = (1, p^2, p^2)^T$. We will see in Example 5.2 that the basis (b_1, b_2, b'_3) is not N -orthogonal.

2.5 p -adic lattices, LVP and CVP

We first recall the definition of a p -adic lattice.

DEFINITION 2.12 (p -ADIC LATTICE, [3]). Let N be a norm on V and let $b_1, \dots, b_m$ be m linearly independent vectors of V . A p -adic lattice in V is the set

$$\mathcal{L}(b_1, \dots, b_m) := \left\{ \sum_{i=1}^m \alpha_i b_i : \alpha_i \in \mathbb{Z}_p \right\} = \sum_{i=1}^m \mathbb{Z}_p b_i$$

of all $\mathbb{Z}_p$ -linear combinations of $b_1, \dots, b_m$. The integers m and n are called respectively the rank and the dimension of the lattice. When $n = m$, the lattice is said to be of full rank.

p -adic lattices are compact subsets of V . Moreover, by [7, p. 72], if $\Omega \subset V$ is compact then $\{N(x - a) : x \in \Omega\}$ is finite for every $a \in V \setminus \Omega$, and $\{N(x - a) : x \in \Omega \setminus \{a\}\}$ is discrete in $\mathbb{R}_{>0}$ for every $a \in \Omega$; this is what makes the two problems below well defined. Finally, if $(b_1, \dots, b_m)$ is an N -orthogonal basis of the subspace $\text{Vect}_{\mathbb{Q}_p}(b_1, \dots, b_m) = \sum_{i=1}^m \mathbb{Q}_p b_i$ of V , it is called an N -orthogonal basis of the lattice $\mathcal{L}(b_1, \dots, b_m)$ [1].

DEFINITION 2.13 (LVP AND CVP, [12]). Let $\mathcal{L} = \mathcal{L}(b_1, \dots, b_m)$ be a p -adic lattice in V , and define recursively $\lambda_1 = \max_{1 \leq i \leq m} N(b_i)$ and $\lambda_{j+1} = \max\{N(v) : v \in \mathcal{L}, N(v) < \lambda_j\}$, so that $\lambda_1 > \lambda_2 > \dots$ and $\lambda_j \rightarrow 0$. The longest vector problem asks for a vector $v \in \mathcal{L}$ with $N(v) = \lambda_2$; given moreover a target vector $t \in V$, the closest vector problem asks for a vector $v \in \mathcal{L}$ with $N(t - v) = \min_{w \in \mathcal{L}} N(t - w)$.

Deng et al. [3] gave deterministic exponential algorithms for LVP and CVP. These problems are therefore natural candidates for cryptographic hardness assumptions. Since both become polynomial-time solvable when an N -orthogonal basis is available [12], understanding the complexity of computing such a basis is essential for assessing the hardness of p -adic lattices constructions. We thus focus on the problem of computing an N -orthogonal basis of V and on its associated computational complexity.

3 A family of norms: $N_{P(\mathbb{F}_p^n)}$

To provide lower complexity bounds for the proofs of the subsequent sections, we define a family of norms whose elements are exponentially hard to distinguish. It is indexed by the vector lines of the projective space $P(\mathbb{F}_p^n)$.

DEFINITION 3.1 (THE FAMILY $N_{P(\mathbb{F}_p^n)}$). For a vector line $x \in P(\mathbb{F}_p^n)$, take $\bar{v} \in x$, $\bar{v} \neq 0$, and lift it to an element $v = \phi(x) \in \mathbb{Z}_p^n$. By construction v is primitive — its coordinates lie in $\mathbb{Z}_p$ and at least one of them belongs to $\mathbb{Z}_p^\times$ — so that it can be completed into a basis $B_x = (v, v_{2,x}, \dots, v_{n,x})$ of $\mathbb{Z}_p^n$. We define the norm N_x on $\mathbb{Q}_p^n$ as follows: if $z = \alpha_1 v + \alpha_2 v_{2,x} + \dots + \alpha_n v_{n,x}$, then

$$N_x(z) := \max(|\alpha_1|_p, p|\alpha_2|_p, \dots, p|\alpha_n|_p),$$

and we set $N_{P(\mathbb{F}_p^n)} := \{N_x : x \in P(\mathbb{F}_p^n)\}$.

Each N_x is indeed an ultrametric norm on $\mathbb{Q}_p^n$: with the notation of Theorem 2.3, it is given in the basis B_x by $c = (1, p, \dots, p)$ and $a_{r,i} = \delta_{r,i}$. In particular all the norms of $N_{P(\mathbb{F}_p^n)}$ take their values in $p^{\mathbb{Z}}$, and they all satisfy $v_{\max} = n$: the family is as far as possible from the incommensurable (easy) case.

LEMMA 3.2. Let $x \in P(\mathbb{F}_p^n)$ and $v = \phi(x)$. The unit ball of N_x is $\Delta_x = \mathbb{Z}_p v + p\mathbb{Z}_p^n$. Consequently, for a primitive vector $z \in \mathbb{Z}_p^n$,

$$N_x(z) = \begin{cases} 1 & \text{if } \bar{z} \in x, \\ p & \text{otherwise,} \end{cases} \quad (2)$$

and the norms of $N_{P(\mathbb{F}_p^n)}$ are pairwise distinct. In particular

$$\text{card}(N_{P(\mathbb{F}_p^n)}) = \frac{p^n - 1}{p - 1}.$$

PROOF. Let $z = \alpha_1 v + \alpha_2 v_{2,x} + \dots + \alpha_n v_{n,x}$ be written in the basis B_x . Then $N_x(z) \leq 1$ if and only if $|\alpha_1|_p \leq 1$ and $p|\alpha_i|_p \leq 1$ for $i \geq 2$, that is, $\alpha_1 \in \mathbb{Z}_p$ and $\alpha_i \in p\mathbb{Z}_p$ for $i \geq 2$; whence the unit ball.

Let now $z \in \mathbb{Z}_p^n$ be primitive, so that all the α_i lie in $\mathbb{Z}_p$. Since $(\bar{v}, \bar{v}_{2,x}, \dots, \bar{v}_{n,x})$ is a basis of $\mathbb{F}_p^n$, we have $\bar{z} \in x$ if and only if $\bar{\alpha}_1 \neq 0$ and $\bar{\alpha}_2 = \dots = \bar{\alpha}_n = 0$. In that case $|\alpha_1|_p = 1$ and $p|\alpha_i|_p \leq 1$ for $i \geq 2$, so $N_x(z) = 1$; otherwise there is some $j \geq 2$ with $\bar{\alpha}_j \neq 0$, so that $p|\alpha_j|_p = p$, while $|\alpha_1|_p \leq 1$ and $p|\alpha_i|_p \leq p$ for every i , whence $N_x(z) = p$. Finally, for $x \neq y$ in $P(\mathbb{F}_p^n)$, the primitive vector $v = \phi(x)$ satisfies $N_x(v) = 1$ (as $\bar{v} \in x$) and $N_y(v) = p$ (as $\bar{v} \notin y$); hence $N_x \neq N_y$, and the cardinality is that of $P(\mathbb{F}_p^n)$. $\square$

PROPOSITION 3.3. Put $m = \frac{p^n-1}{p-1}$. For any $s \leq m-2$ nonzero vectors $w_1, \dots, w_s$ of $\mathbb{Q}_p^n$, there exist two distinct norms of $N_{P(\mathbb{F}_p^n)}$ which coincide on each of them. Consequently, at least $m-1 = p + p^2 + \dots + p^{n-1}$ vectors are needed to tell apart all the norms of $N_{P(\mathbb{F}_p^n)}$, and this bound is sharp.

PROOF. For $i \in \llbracket 1, s \rrbracket$, write $w_i = p^{k_i} w_{i,0}$ with $w_{i,0} \in \mathbb{Z}_p^n$ primitive, and let $x_i \in P(\mathbb{F}_p^n)$ be the vector line defined by $\overline{w_{i,0}}$. By Lemma 3.2 there are m distinct norms in $N_{P(\mathbb{F}_p^n)}$, and the set $\{x_1, \dots, x_s\}$ has at most $m-2$ elements, so there are two distinct lines $y_1 \neq y_2 \in P(\mathbb{F}_p^n) \setminus \{x_1, \dots, x_s\}$. For every i , $\overline{w_{i,0}}$ belongs neither to y_1 nor to y_2 , so that $N_{y_1}(w_{i,0}) = N_{y_2}(w_{i,0}) = p$ by (2); by homogeneity $N_{y_1}(w_i) = p^{-k_i} N_{y_1}(w_{i,0}) = p^{-k_i} N_{y_2}(w_{i,0}) = N_{y_2}(w_i)$, and both norms coincide on the w_i 's. Conversely, $m-1$ vectors attached to $m-1$ pairwise distinct projective lines do separate the family: given $y_1 \neq y_2$, at least one of these two lines is represented, say $y_1 = x_i$, and then $N_{y_1}(w_{i,0}) = 1 \neq p = N_{y_2}(w_{i,0})$. $\square$

4 Orthogonal basis computation

Recall that knowing an (A, c) -representation of the norm N is equivalent to knowing an N -orthogonal basis, by Theorem 2.3 and its proof; in that model, orthogonalisation — hence LVP and CVP, by [12] — is polynomial. We therefore assume in this section that only a norm oracle is available.

4.1 Lower bound: no polynomial algorithm

THEOREM 4.1 (LOWER BOUND ORTHOGONALIZATION). Any algorithm (deterministic, or randomised with success probability $\geq 2/3$) which reconstructs an (A, c) -representation of N from a norm oracle performs $\Omega(p^{v_{\max}-1})$ oracle calls in the worst case. In particular, for $v_{\max} = n: \Omega(p^{n-1})$.

The idea is that, on a given query, all the norms of $N_{P(\mathbb{F}_p^n)}$ return the same answer, except possibly the one whose secret direction is that of the queried vector: each query rules out at most one candidate among the $\frac{p^n-1}{p-1}$ possible ones.

PROOF. Assume first $v_{\max} = n$ and let $K = \text{card}(P(\mathbb{F}_p^n)) = \frac{p^n-1}{p-1}$, so that $N_{P(\mathbb{F}_p^n)}$ consists of K distinct norms, all with $v_{\max} = n$. Let $\mathcal{A}$ be a deterministic algorithm performing T queries, and run it against the adversary that answers p^{1-k} to a query $w = p^k w_0$ with $w_0 \in \mathbb{Z}_p^n$ primitive. By (2), this transcript is exactly the one produced by N_y for every line y outside the set of the (at most T) lines spanned by the reductions of the queried vectors; on all these instances $\mathcal{A}$ behaves identically and returns the same pair (A, c) , which, determining the norm it represents, is correct for at most one of them. Hence $\mathcal{A}$ is correct on at most $T+1$ of the K lines, and its success probability under the uniform distribution on $P(\mathbb{F}_p^n)$ is at most $(T+1)/K$: a deterministic algorithm needs $T \geq K-1$, which is Proposition 3.3 again, and in the randomised case Yao's principle gives $(T+1)/K \geq 2/3$, that is, $T = \Omega(\frac{p^n-1}{p-1}) = \Omega(p^{n-1})$.

For an arbitrary $v_{\max} \leq n$, apply this construction inside the subspace spanned by the first v coordinates and set $N(z) := \max(N_x(z_1, \dots, z_v), c_{v+1}|z_{v+1}|_p, \dots, c_n|z_n|_p)$, where $c_{v+1}, \dots, c_n$ and 1 are pairwise incommensurable: the last $n-v$ directions are orthogonal to everything by Proposition 2.8 and are

identified at no cost, the multiplicity of the class $p^{\mathbb{Z}}$ is $v = v_{\max}$, and the problem restricts to the v -dimensional instance above. $\square$

4.2 The algorithm

To lighten the notation, we write $\gamma_i = N(b_i)$ where (b_i) is a basis of V . Here is the fundamental criterion used by the algorithm.

LEMMA 4.2. Let (b_i) be a basis of V endowed with a norm N such that, for every i ,

$$N(b_i) = \min_{\delta \in W_i} N(b_i - \delta), \quad W_i = \text{Vect}(b_j)_{j \neq i}.$$

Then (b_i) is N -orthogonal.

PROOF. The minimum exists: by the ultrametric equality, $\delta \in W_i$ and $N(b_i - \delta) < N(b_i) = \gamma_i$ force $N(\delta) = \gamma_i$, so that the search space is not the whole of W_i (which is not compact) but $W_i \cap B_{\leq \gamma_i}$, a p -adic lattice, hence compact. Assume now that there is $v = \sum \alpha_i b_i \in V$ with $N(v) < \rho := \max_i |\alpha_i|_p \gamma_i$ (the inequality $\leq$ being automatic), let i_0 achieve the maximum, so that $\alpha_{i_0} \neq 0$, and put $\delta = -\sum_{j \neq i_0} \frac{\alpha_j}{\alpha_{i_0}} b_j \in W_{i_0}$. Then $N(b_{i_0} - \delta) = |\alpha_{i_0}|_p^{-1} N(v) < |\alpha_{i_0}|_p^{-1} \rho = \gamma_{i_0}$, contradicting the minimality of $N(b_{i_0})$. $\square$

In other words, local descent implies global optimality: no random restart is needed, in contrast with real gradient descent — an important point, since the orthogonalisation algorithm with an oracle is exactly a discrete p -adic descent: one starts from a defect of minimality in the norms of the b_i and corrects it by finding $\delta \in W_i$ making $N(b_i - \delta)$ minimal. Such a correction is exactly an instance of CVP, the ultrametric inequality restricting the search space to the ball $W_i \cap B_{\leq N(b_i)}$, a finitely generated $\mathbb{Z}_p$ -module; it is this geometric reduction of a vector problem to a lattice problem that gives a transparent intuition for the polynomial equivalence, already known, between orthogonalisation and CVP.

The next lemma is the engine of the algorithm: it makes the descent test finite and confines it to a single commensurability block.

LEMMA 4.3. Let $(e_1, \dots, e_{k-1})$ be an N -orthogonal family, $\gamma_j = N(e_j)$, and let $x \notin W = \text{Vect}(e_1, \dots, e_{k-1})$ with $\gamma = N(x)$. Put

$$J = \left\{ j < k : \frac{\gamma_j}{\gamma} \in p^{\mathbb{Z}} \right\}, \quad k_j := \log_p(\gamma_j/\gamma) \quad (j \in J),$$

so that $N(p^{k_j} e_j) = N(x)$ for $j \in J$. Then $(e_1, \dots, e_{k-1}, x)$ is N -orthogonal if and only if

$$N\left(x - \sum_{j \in J} \beta_j p^{k_j} e_j\right) = \gamma \quad \text{for all } \beta_j \text{'s in } \{0, 1, \dots, p-1\}.$$

Moreover $\text{card } J \leq v_\gamma \leq v_{\max}$.

PROOF. By Lemma 2.10, the family $(e_1, \dots, e_{k-1}, x)$ is N -orthogonal if and only if each of its commensurability blocks is. The blocks that do not contain x are subfamilies of $(e_1, \dots, e_{k-1})$, hence are orthogonal; the block containing x is $\{x\} \cup \{e_j : j \in J\}$, which after the normalisation $e_j \mapsto p^{k_j} e_j$ — a rescaling that changes neither the block nor its orthogonality — consists of vectors of norm γ . By Lemma 2.7, that block is orthogonal if and only if $N(\bar{\alpha}_0 x + \sum_{j \in J} \bar{\alpha}_j p^{k_j} e_j) = \gamma$ for all $\bar{\alpha} \in \{0, \dots, p-1\}^{J \cup \{0\}}$ not all zero; if $\bar{\alpha}_0 = 0$ this holds automatically, the e_j being already orthogonal, and if $\bar{\alpha}_0 \neq 0$, dividing by the unit $\bar{\alpha}_0$ brings us back to the stated form. Finally, the same

Algorithm 2: DESCEND: orthogonalisation with an oracle

Input: a norm oracle for N on V , a basis $B = (b_1, \dots, b_n)$

Output: an N -orthogonal basis $(e_1, \dots, e_n)$ of V

```
1  $E \leftarrow \emptyset; W \leftarrow \{0\};$ 
2 for  $k = 1, \dots, n$  do
3    $x \leftarrow \arg \min_{b_j \notin W} N(b_j); \gamma \leftarrow N(x);$ 
   // descent to a best approximation (CVP)
4   while true do
5      $J \leftarrow \{j < k : \gamma_j / \gamma \in p^{\mathbb{Z}}\};$ 
6      $k_j \leftarrow \log_p(\gamma_j / \gamma)$  for  $j \in J$ ;
7      $found \leftarrow \text{false};$ 
8     foreach  $\beta \in \{0, \dots, p-1\}^J \setminus \{0\}$  do
9        $y \leftarrow x - \sum_{j \in J} \beta_j p^{k_j} e_j;$ 
10      if  $N(y) < \gamma$  then
11         $x \leftarrow y; \gamma \leftarrow N(y); found \leftarrow \text{true}; \text{break};$ 
12    if not  $found$  then break //  $x$  is optimal;
13     $e_k \leftarrow x; \gamma_k \leftarrow \gamma; W \leftarrow W \oplus \mathbb{Q}_p \cdot e_k;$ 
14 return  $E = (e_1, \dots, e_n);$ 
```

criterion applied to the family $(p^{k_j} e_j)_{j \in J}$ shows that every nonzero $\mathbb{F}_p$ -linear combination of these vectors has norm γ by orthogonality. Therefore, their images in $\text{gr}_\gamma = B_{\leq \gamma} / B_{< \gamma}$ are $\mathbb{F}_p$ -independent, thus $\text{card } J \leq v_\gamma$. $\square$

We can now build in Algorithm 2 a recursive algorithm to compute an N -orthogonal basis $(e_1, \dots, e_n)$ out of a basis $(b_1, \dots, b_n)$ of V . At step k , assuming $(e_1, \dots, e_{k-1})$ orthogonal, it checks whether the norm of $x := b_k$ can be made to drop by subtracting an $\mathbb{F}_p$ -linear combination of those e_j that are commensurable with it; if so it replaces x by the new vector (it goes down one “floor”) and starts again, until no further descent is possible, i.e. until the global minimum has been found. When no $\mathbb{F}_p$ -linear combination makes the norm drop, $e_k := x$ is orthogonal to $(e_1, \dots, e_{k-1})$ by Lemma 4.3.

To evaluate the complexity of this algorithm, we must bound the number of times the norm of a vector can drop. At the k -th iteration, let $x_k^{(0)}$ denote the vector selected at step 3 of the algorithm, γ its norm and let W_{k-1} be the subspace spanned by the first $k-1$ orthogonalised vectors. Since $x_k^{(0)} \notin W_{k-1}$ and W_{k-1} is closed, the distance between $x_k^{(0)}$ and W_{k-1} , $d_k := \min_{w \in W_{k-1}} N(x_k^{(0)} - w)$ is strictly positive. Recall that the set of values taken by N is contained in $\Gamma = \bigcup_{r=1}^n c_r p^{\mathbb{Z}} \cup \{0\}$. For each r , the class $c_r p^{\mathbb{Z}}$ contains at most one element in the interval $[\gamma/p, \gamma]$, since two consecutive elements of this class differ by a factor p . Hence the norm can take at most n distinct values on an interval $[\gamma/p, \gamma]$. Since each update $x \leftarrow x - w$ makes the norm drop strictly, after at most n updates the norm is at most γ/p . Repeating this argument, after t updates the working vector $x_k^{(t)}$ satisfies $N(x_k^{(t)}) \leq N(x_k^{(0)}) p^{-\lfloor t/n \rfloor}$. On the other hand, every working vector is of the form $x_k^{(0)} - w$, $w \in W_{k-1}$, so its norm is at least d_k . Therefore the number of updates is at most $n \lceil \log_p(N(x_k^{(0)})/d_k) \rceil$.

We thus define the *geometric conditioning* of the instance as the maximal number of descents,

$$D = \max_{2 \leq k \leq n} n \left\lceil \log_p \left(\frac{N(x_k^{(0)})}{d_k} \right) \right\rceil,$$

a parameter which is finite and fixed by the input.

PROPOSITION 4.4 (TERMINATION, COMPLEXITY, CORRECTNESS). *Algorithm 2 terminates after finitely many steps, with a query complexity of*

$$\# \text{oracle calls} = O(n(1 + D) \cdot p^{v_{\max}}).$$

Moreover, at the end of iteration k , the family $(e_1, \dots, e_k)$ is N -orthogonal.

PROOF. *Termination and complexity.* By definition of D , for each vector $k \leq n$ the algorithm finds at most D successive combinations that decrease the norm, hence performs at most $1 + D$ complete explorations of a level (the last one certifying optimality). During one exploration, the space of coefficients $\mathbb{F}_p^J$ has size at most $p^{v_{\max}}$ by Lemma 4.3, so a level costs $O(p^{v_{\max}})$ oracle calls; summing over the n basis vectors gives the announced complexity, and in particular termination.

Correctness. Assume by induction that $(e_1, \dots, e_{k-1})$ is orthogonal. At step k the loop stops when the search over J fails, which by Lemma 4.3 means exactly that $(e_1, \dots, e_{k-1}, e_k)$ is N -orthogonal — geometrically, by Lemma 4.2, that e_k is at minimal distance from W_{k-1} . $\square$

Comparing with the lower bound $\Omega(p^{v_{\max}-1})$, the algorithm is optimal up to a factor $\text{poly}(n, D) \cdot p$; its arithmetic cost is polynomial, only $\mathbb{Z}$ -linear combinations being performed. Certifying that a basis is N -orthogonal, on the other hand, requires solving the orthogonality decision problem, which is the subject of the next section.

EXAMPLE 4.5. *Take again the norm $N(v) = \max(\pi|v_1|_p, \sqrt{2}|v_1 + v_2|_p, |v_1 + v_2 + v_3|_p)$, assuming $p \geq 5$, and the basis B of Example 2.11. Since b_1 and b_2 belong to distinct commensurability classes (generated by 1 and π), they are N -orthogonal from the start and we set $e_1 = b_1$, $e_2 = b_2$. The vector b_3 , on the other hand, is commensurable with e_2 : Algorithm 2 first normalises it as $b'_3 = p^2 b_3 = (1, p^2, p^2)^T$, of norm π , then kills the dependency in the residual space of the class π by a cancellation modulo p ,*

$$x = b'_3 - e_2 = (-p^2, p^2 - p^3, -p)^T,$$

whose norm drops strictly to $N(x) = \max(\pi p^{-2}, \sqrt{2} p^{-3}, p^{-1}) = p^{-1}$ (because $p \geq 5$): the vector x has just changed commensurability class and joined that of e_1 . A second elimination, now in the class of 1, gives

$$e_3 = x + p^3 e_1 + p^2 e_2 = (p^3 + p^4, p^2 - p^3 + p^5 + p^7, p^3 + 2p^4)^T,$$

of norm $\max(\pi p^{-3}, \sqrt{2} p^{-2}, p^{-2}) = \sqrt{2} p^{-2}$: the vector e_3 has reached the class of $\sqrt{2}$, which was vacant, so that (e_1, e_2, e_3) is an N -orthogonal basis by Remark 2.9(iii). For readability we used the representative -1 of $p-1$ modulo p . However by the ultrametric inequality, it does not affect whether the norm of the corresponding combination drops or not.

Finally, it is clear that D depends on the chosen norm: it is the algorithmic conditioning of the instance, and depending on N it can be arbitrarily large. This exhibits the two distinct parameters that govern the cost. The multiplicity $v_{\max}$ of the largest commensurability class yields an intrinsic barrier $\Omega(p^{v_{\max}-1})$ that no algorithm can cross, whereas the geometric conditioning D is exactly what makes a uniform bound fail, as in [4]. Since D is linear in n and logarithmic in the accuracy needed to describe the instance, it is not a plausible security parameter, while $p^{v_{\max}}$ is an unconditional one.

5 Deciding orthogonality

5.1 The residual commensurability criterion

In this subsection, suppose that an (A, c) -representation of the norm is known. Let $B = (b_1, \dots, b_m)$ be a family of nonzero vectors of V and let $B_1, \dots, B_k$ ($1 \leq k \leq m$) be its commensurability classes. In each block B_i we may normalise every vector, $b'_j = \alpha_j b_j$, so that all the b'_j have the same norm $\rho_i > 0$; we define $\tilde{M}_i$, the residual matrix with coefficients in $\mathbb{F}_p$ containing the leading digits of the dominant $L_r(b'_j)$, as follows.

The columns of $\tilde{M}_i$ are indexed by the indices j such that $b_j \in B_i$. The rows of $\tilde{M}_i$ are indexed by the indices r such that $c_r |L_r(b_j)|_p = \rho_i$ for some $b_j \in B_i$.

For such a couple of r and j , $|L_r(b_j)|_p = \frac{\rho_i}{c_r}$. As $|\cdot|_p$ takes its values in $p^{\mathbb{Z}} \cup \{0\}$, this forces $\rho_i/c_r \in p^{\mathbb{Z}}$, say $\rho_i/c_r = p^{-d_r^{(i)}}$ with $d_r^{(i)} \in \mathbb{Z}$. In that case, one can write $L_r(b_j) = a_0 p^{d_r^{(i)}} + \dots$ and $a_0 = L_r(b_j) p^{-d_r^{(i)}} \bmod p \in \mathbb{F}_p$. For the same r and other j 's of the same commensurability class, $|L_r(b_j)|_p \leq \frac{\rho_i}{c_r} = p^{-d_r^{(i)}}$, thus, $L_r(b_j) p^{-d_r^{(i)}} \bmod p \in \mathbb{F}_p$ and is equal to 0 when $c_r |L_r(b_j)|_p < \rho_i$.

As such, we can define $\tilde{M}_i$, a matrix with coefficients in $\mathbb{F}_p$, as follows: for j such that $b_j \in B_i$ and for r such that $c_r |L_r(b_j)|_p$ reaches ρ_i ,

$$(\tilde{M}_i)_{r,j} := L_r(b_j) p^{-d_r^{(i)}} \bmod p \in \mathbb{F}_p.$$

THEOREM 5.1 (RESIDUAL COMMENSURABILITY CRITERION). *Let N be a norm on V and let $b_1, \dots, b_m$ be nonzero vectors of V . Let $B_1, \dots, B_k$ be their commensurability classes and $\tilde{M}_1, \dots, \tilde{M}_k$ the associated residual matrices of leading digits. Then the family is N -orthogonal if and only if $\ker \tilde{M}_i = \{0\}$ over $\mathbb{F}_p$ for every block B_i .*

PROOF. By Lemma 2.10 the family is N -orthogonal if and only if each block is, so we may work inside a single block of commensurability, of size l say; and the normalisation $b'_j := \alpha_j b_j$ with $\alpha_j \in \mathbb{Q}_p^*$ does not affect orthogonality — the factor is absorbed by the coefficients of the linear combination — so we may assume that all the b_j have norm ρ . By Lemma 2.7, the block is then N -orthogonal if and only if $N(b) = \rho$ for every $b = \sum_{j=1}^l \tilde{\alpha}_j b_j$ with $\tilde{\alpha} \in \{0, \dots, p-1\}^l \setminus \{0\}$.

Let us assume that our block is not N -orthogonal. Let $\tilde{\alpha} \in \{0, \dots, p-1\}^l \setminus \{0\}$ be a corresponding combination with $N(\sum_{j=1}^l \tilde{\alpha}_j b_j) < \rho$.

Algorithm 3: Residual commensurability with (A, c)

Input: a family B of $m \geq 2$ vectors b_j of size n , a diagonal representation (A, c) of a norm N
Output: true if B is N -orthogonal, false otherwise

- 1 compute the $k \leq m$ commensurability classes with Algorithm 1;
- 2 **if** card *Classes* = m **then return** true // pairwise incommensurable;
- 3 **foreach** class B_i of $l_i \geq 2$ vectors, of common norm ρ_i **do**
- 4 build $\tilde{M}_i$, having n_i rows;
- 5 **if** $n_i > l_i$ or $\ker \tilde{M}_i \neq \{0\}$ **then return** false;
- 6 **return** true;

Then $\max_r c_r |L_r(\sum_{j=1}^l \tilde{\alpha}_j b_j)|_p < \rho$. As such, for any r ,

$$\text{val} \left(L_r \left(\sum_{j=1}^l \tilde{\alpha}_j b_j \right) \right) > -\log_p \left(\frac{\rho}{c_r} \right) = d_r.$$

Consequently, for a given r ,

$$\text{val} \left(\sum_{j=1}^l p^{-d_r} L_r(b_j) \tilde{\alpha}_j \right) > 0$$

and for every j , $\text{val} (p^{-d_r} L_r(b_j) \tilde{\alpha}_j) \geq 0$. By the ultrametric inequality, it means that in $\mathbb{F}_p$, $\sum_{j=1}^l (p^{-d_r} L_r(b_j) \tilde{\alpha}_j \bmod p) = 0$ and thus

$$\sum_{j=1}^l \tilde{M}_{r,j} \tilde{\alpha}_j = 0.$$

This means that $\tilde{\alpha}$ is a non-zero element of $\ker \tilde{M}$, which proves the first implication.

Conversely, let $\tilde{\alpha}$ be a non-zero element of $\ker \tilde{M}$ and let r be the index of a row of $\tilde{M}$.

We get that $\sum_{j=1}^l \tilde{M}_{r,j} \tilde{\alpha}_j = 0$. As before, for every j , $\text{val} (p^{-d_r} L_r(b_j) \tilde{\alpha}_j) \geq 0$ and in addition, $\sum_{j=1}^l (p^{-d_r} L_r(b_j) \tilde{\alpha}_j \bmod p) = 0$. This means that $(\sum_{j=1}^l (p^{-d_r} L_r(b_j) \tilde{\alpha}_j)) \bmod p = 0$, and thus $\text{val} (\sum_{j=1}^l p^{-d_r} L_r(b_j) \tilde{\alpha}_j) > 0$. Consequently, for any r ,

$$\text{val} \left(L_r \left(\sum_{j=1}^l \tilde{\alpha}_j b_j \right) \right) > -\log_p \left(\frac{\rho}{c_r} \right) = d_r,$$

and finally, $\max_r c_r |L_r(\sum_{j=1}^l \tilde{\alpha}_j b_j)|_p < \rho$, which concludes the proof. $\square$

Theorem 5.1 gives an algorithm to test orthogonality: compute the commensurability classes, build the matrix $\tilde{M}_i$ of each block from the formula above — reading off the valuations and the leading digits of the $L_r(b_j)$, which costs $O(nl)$ basic operations for a block of size l — and compute its kernel.

One optimisation can be added: a matrix with strictly more columns than rows already has a non-trivial kernel. As such, kernel computation can be avoided for the so-called ‘wide’ matrices. The resulting Algorithm is presented as Algorithm 3.

Let us study the complexity of Algorithm 3: in the worst case all the vectors are commensurable and the kernel computation cannot be split into smaller blocks and dominates the running time,

which gives $O(nm^2)$ elementary operations. It becomes linear in the favourable cases.

EXAMPLE 5.2. Take again the norm and the basis of Example 2.11, where b_1 is alone in its commensurability class while b_2 and b_3 are commensurable with each other. The last vector is replaced by $b'_3 = p^2 b_3 = (1, p^2, p^2)^T$. We get that $\rho_2 = N(b_2) = N(b'_3) = \pi$. This value is reached only by $\pi|L_1(b_2)|_p = \pi|L_1(b'_3)|_p$. The $c_2|L_2(b_j)|_p$ and $c_3|L_3(b_j)|_p$ are not even commensurable with π . As such, $\bar{M}_2$ has one row and two columns, being equal to $\bar{M} = \begin{pmatrix} 1 & 1 \end{pmatrix}$. The kernel vector $(1, -1)$ even exhibits the cancellation, $b'_3 - b_2$, which is precisely the one that made the norm drop in Example 4.5.

REMARK 5.3. (i) For the sup norm, all vectors can be normalised so that their norm becomes 1. The matrix $\bar{M}$ is then simply the matrix of the normalised vectors mod p , and the family is orthogonal if and only if $\bar{M}$ has maximal rank.

(ii) This criterion requires knowing a diagonal representation of the norm: both the c_r and the L_r are needed.

5.2 With a norm oracle

The previous subsection decides orthogonality when an (A, c) -representation of the norm is available. With only a norm oracle at hand, kernel computations are out of reach. Given a block of l commensurable vectors, all of norm ρ — the commensurability classes themselves are computed with Algorithm 1 — Lemma 2.7 tells us it is enough to check whether $N(\sum_{i=1}^l \alpha_i b_i) = \max_i |\alpha_i|_p \rho$ for every $\alpha_1, \dots, \alpha_l \in \{0, \dots, p-1\}$, not all zero. This naively gives $p^l - 1$ combinations. However, if $v = \sum_{i=1}^l \alpha_i b_i$ makes the norm drop, so does αv for every $\alpha \in \{1, \dots, p-1\}$, since $N(\alpha v) = |\alpha|_p N(v) = N(v)$; working in the projective space avoids these $p-1$ repetitions and leaves

$$\frac{p^l - 1}{p - 1}$$

norm computations to perform. This remains exponential in l , hence not very practical. It is worth noting that it is exactly the count obtained by Duan and Wu for the worst-case query complexity of the LVP in a lattice of rank l [4].

We write $l_{\max}$ for the number of vectors in the largest commensurability class; it can be any integer between 1 and n , depending entirely on the family being tested, and is known as soon as the classes have been computed. This leads us to the following theorem.

THEOREM 5.4 (LOWER BOUND DECISION). Any algorithm (deterministic, or randomised with success probability $\geq 2/3$) that decides whether a basis B is N -orthogonal performs $\Omega(p^{l_{\max}-1})$ oracle calls in the worst case. In particular, if $l_{\max} = n$: $\Omega(p^{n-1})$.

PROOF. Let us place ourselves in the worst case $l_{\max} = n$ and take for $\mathcal{B} = (b_1, \dots, b_n)$ a $\mathbb{Z}_p$ -basis of $\mathbb{Z}_p^n$, so that the reductions $\bar{b}_i$ form a basis of $\mathbb{F}_p^n$. We build oracles N_0 and N_x that the algorithm cannot tell apart without $\Omega(p^{n-1})$ queries, and such that $\mathcal{B}$ is N_0 -orthogonal but not N_x -orthogonal.

For the YES instance, define $N_0(b) = p \max_{1 \leq i \leq n} |\alpha_i|_p$, where the α_i are the coordinates of b in $\mathcal{B}$: then $N_0(b_i) = p$, and every primitive linear combination $b = \sum \alpha_i b_i$ satisfies $\max_i |\alpha_i|_p = 1$, so that $N_0(b) = p$. No combination makes the norm drop and $\mathcal{B}$ is N_0 -orthogonal. For the NO instances, pick a secret direction $x \in P(\mathbb{F}_p^n)$

Algorithm 4: Residual commensurability with oracle

Input: a family B of $m \geq 2$ vectors b_j of size n ; a norm oracle for N

Output: true if B is N -orthogonal, false otherwise

```

1 compute the  $k \leq m$  commensurability classes with
  Algorithm 1;
2 if card Classes =  $m$  then return true          // pairwise
  incommensurable;
3 foreach class  $B_i$  of  $l_i \geq 2$  vectors, of common norm  $\rho_i$  do
4   foreach combination  $v$  of  $B_i$  with coefficients mod  $p$ , up
    to projective equivalence do
5     if  $N(v) < \rho_i$  then return false;
6 return true;
```

such that $\bar{b}_i \notin x$ for every i ; there are $K = \frac{p^n - 1}{p - 1} - n \geq 1$ such lines, and by (2) the norm $N_x \in N_{P(\mathbb{F}_p^n)}$ satisfies $N_x(b_i) = p$ for every i , while the linear combination pointing in the direction of x sees its norm drop to $1 < p$: the basis $\mathcal{B}$ is not N_x -orthogonal.

Now, for $w = p^k w_0$ with w_0 primitive, $N_0(w) = p^{1-k}$, and $N_x(w) = p^{1-k}$ as well unless $\bar{w}_0 \in x$. Run a deterministic algorithm making T queries against the adversary of Theorem 4.1, answering p^{1-k} : the transcript is consistent with N_0 and with N_x for every one of the $\geq K - T$ admissible lines x not spanned by the reduction of a queried vector. If the algorithm answers “orthogonal” it is wrong on all of these; if it answers “not orthogonal” it is wrong on N_0 . Correctness on every instance thus forces $T \geq K = \Omega(p^{n-1})$. For the randomised case, take the distribution that picks N_0 with probability $1/2$, and otherwise a uniform N_x among the K admissible ones: a deterministic algorithm making T queries then errs with probability at least $\frac{1}{2} \min(1, \frac{K-T}{K})$, so an error probability $< 1/3$ forces $T > K/3$, and Yao’s principle concludes. Finally, running the same construction inside the largest commensurability block, of size $l_{\max}$, and completing with pairwise incommensurable directions (which are orthogonal to everything by Proposition 2.8 and cost nothing to handle), gives $\Omega(p^{l_{\max}-1})$. $\square$

The matching algorithm is written as Algorithm 4.

Inside each block of l_i commensurable vectors of common norm ρ_i , it enumerates the $\frac{p^{l_i}-1}{p-1}$ projective combinations with coefficients modulo p and returns false as soon as one of them has norm $< \rho_i$. Its complexity is then

$$O\left(m + \sum_i \frac{p^{l_i} - 1}{p - 1}\right).$$

In the worst case, the largest block alone contributes $\Theta\left(\frac{p^{l_{\max}} - 1}{p - 1}\right) = \Theta(p^{l_{\max}-1})$ oracle queries. This is precisely the contribution captured by our lower bound in Theorem 5.4. When there is a single block, this gives a matching bound

$$\Theta(p^{l_{\max}-1}).$$

COROLLARY 5.5. Relative to a norm oracle, deciding N -orthogonality is in co-NP but not in NP.

PROOF. A negative instance has a short certificate: a nonzero $\bar{\alpha} \in \mathbb{F}_p^l$ together with the index of the block, which is checked with

a single oracle call by Lemma 2.7; hence the problem is in co-NP. Conversely, assume it were in NP and consider the instance $(\mathcal{B}, N_0)$ of the proof of Theorem 5.4: a verifier reading a certificate and making T queries accepts it, and the same certificate is accepted verbatim for every norm N_x consistent with the transcript, of which there are at least $K - T$. For T polynomial in n and $\log p$, such an N_x exists and is a negative instance, a contradiction. $\square$

6 Conclusion and further work

The hardness of the orthogonalisation and decision problems comes above all from the representation of the norm and from commensurability: when all the vectors are pairwise incommensurable everything becomes polynomial, whereas the complexity is exponential in $v_{\max}$ (resp. $l_{\max}$) otherwise. Moreover, the proofs of Theorems 4.1 and 5.4 show that these problems amount to identifying a hidden subgroup — here a hidden line of $\mathbb{F}_p^n$ — through membership queries, a setting whose hardness is well understood [5]: it does admit Grover-type quantum speed-ups, in $p^{(n-1)/2}$, but nothing better is known to date and the gain remains marginal. To escape the exponential regime one has to change the oracle model; an *argmax* oracle returning both the value of the norm and an index r achieving the maximum, for instance, would give information on the linear forms directly, with no exhaustive search.

We therefore leave the following question open: what is the hardness of these problems in the intermediate case where the norm is given neither by a diagonal form (the easy case) nor by a black-box oracle (the hard case), but by a public representation designed to hide a diagonal form? In other words, is it possible to build a representation of a norm N that reveals no N -orthogonal basis? This opens the way to new asymmetric cryptographic constructions.

References

- [1] Yingpu Deng. “On p -adic Gram–Schmidt Orthogonalization Process”. In: *Frontiers of Mathematics* 20.2 (2025), pp. 299–311.
- [2] Yingpu Deng, Lixia Luo, Yanbin Pan, Zhaonan Wang, and Guanju Xiao. “Public-key cryptosystems and signature schemes from p -adic lattices”. In: *p-Adic Numbers, Ultrametric Analysis and Applications* 16.1 (2024), pp. 23–42.
- [3] Yingpu Deng, Lixia Luo, Yanbin Pan, and Guanju Xiao. “On some computational problems in local fields”. In: *Journal of Systems Science and Complexity* 35.3 (2018), pp. 1191–1200.
- [4] Zhefan Duan and Huawei Wu. “Norm-Query Complexity of Algorithmic Problems in Finite-Dimensional p -adic Normed Spaces”. In: *arXiv preprint arXiv:2608.22084* (2026).
- [5] Simone Dutto, Pietro Mercuri, Nadir Murru, and Lorenzo Romano. “A survey about Hidden Subgroup Problem from a mathematical and cryptographic perspective”. In: *arXiv preprint arXiv:2512.02087* (2025).
- [6] National Institute of Standards and Technology. *Post-Quantum Cryptography*. last accessed 2026-08-05.
- [7] Alain M Robert. *A course in p -adic analysis*. Vol. 198. Springer Science & Business Media, 2000.
- [8] Jean Pierre Serre. *Corps locaux*. Institut de mathématiques de l’université de Nancago, 1968.
- [9] Daniel R. Simon. *A Polynomial-Time Quantum Algorithm for the Dihedral Coset Problem*. Cryptology ePrint Archive, Paper 2026/1591. 2026.
- [10] André Weil. *Basic number theory*. Vol. 1974. Springer, 1967.
- [11] Chi Zhang and Yingpu Deng. “An Explicit Construction of Orthogonal Basis in p -adic Fields”. In: *arXiv preprint arXiv:2410.17982* (2024).
- [12] Chi Zhang, Yingpu Deng, and Zhaonan Wang. “Norm orthogonal bases and invariants of p -adic lattices”. In: *Linear Algebra and its Applications* (2025).
- [13] Chi Zhang and Mingqian Yao. “Notes on the LVP and the CVP in p -adic Fields”. In: *arXiv preprint arXiv:2512.24207* (2026).